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Distance-Based Analysis of Ordinal Data and Ordinal Time Series

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ABSTRACT
The dissimilarity of ordinal categories can be expressed with a distance measure. A unified approach relying on expected distances is proposed to obtain well-interpretable measures of location, dispersion, or symmetry of random variables, as well as measures of serial dependence within a given process. For special types of distance, these analytic tools lead to known approaches for ordinal or real-valued random variables. We also analyze the sample counterparts of the proposed measures and derive asymptotic results for practically important cases in ordinal data and time series analysis. Two real applications about the economic situation in Germany and the credit rating of European countries are presented. Supplementary materials for this article are available online.

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1. Introduction

This article is about the analysis of *ordinal samples* $X_1, \ldots, X_n$ having the ordered categorical range $\mathcal{S} = \{s_0, \ldots, s_m\}$ with $s_0 < \cdots < s_m$. $X_1, \ldots, X_n$ are assumed to be either independent and identically distributed (iid), or to origin from some stochastic process $(X_t)_{\mathbb{Z}}$. In the first case, we refer to the realizations $x_1, \ldots, x_n$ as *ordinal data*, and in the second case as an *ordinal time series*. Mathematically, we can interprete the iid-case as being included in stochastic-process approach; but since iid-applications might not even include a time aspect, we treat these two scenarios separately here.

Ordinal random variables X are closely related to count random variables having the bounded range $\{0, \ldots, m\}$. This can be seen by introducing the *rank-count* variable I with bounded range $\{0, \ldots, m\}$, which is defined by the relation $X = s_I$. Thus, I counts by how many categories X departs from its lowest category. Similarly, the count process $(I_t)_{\mathbb{Z}}$ generates the ordinal process $(X_t)_{\mathbb{Z}}$ by setting $X_t = s_{I_t}$, and consequently, $(X_t)_{\mathbb{Z}}$ inherits distributional properties from $(I_t)_{\mathbb{Z}}$ like stationarity, mixing properties as well as marginal and joint probabilities (Weiß 2018). In particular, we have

$$\begin{aligned} P(X_t = s_k) &= P(I_t = k), \\ P(X_t = s_k, X_{t-h} = s_l) &= P(I_t = k, I_{t-h} = l). \end{aligned}$$

This *rank-count duality* is related (in a sense) to the *latent-variable approach* for ordinal random variables (Agresti 2010), but the rank counts are neither continuous-valued nor unobservable. The latent-variable approach is often used as a way of obtaining a parametric model for ordinal data. Note that there are only few proposals of parametric distribution families for ordinal random variables in the literature, like the "λ-μ-distribution" proposed by Kvålseth (2011b) and the "Lehmann approach" for parametrically modifying ordinal distributions (Lehmann 1953). So it is good to know that the rank-count approach offers a further way of defining parametric models for ordinal random variables and also ordinal processes. As an example, Klein and Doll (2018) used the binomial distribution for generating skewed ordinal random variables. Using a model for time series of bounded counts instead (Weiß 2018), one can generate an ordinal process.

In this work, we consider distance functions defined on the ordinal range $\mathcal{S}$ of the considered random variables, and we define measures of marginal or dependence properties based on expectations of such distances. The advantages of this unified approach are a great flexibility with respect to tailor-made distances (motivated, e.g., by a specific application scenario), a unique interpretation as well as a universal applicability, since distances can be defined for any kind of data, not only for ordinal categories. Actually, distance-based approaches are commonly used in statistical analysis. For instance, Székely, Rizzo, and Bakirov (2007) measure serial dependence in real-valued processes by analyzing distances between characteristic functions, or Kvålseth (2011a) defines measures of ordinal variation based on distances between cumulative distribution functions (CDFs). An essential difference between these approaches and the present work is that our starting point are distances defined on the ordinal range $\mathcal{S}$ itself, but not between CDFs or other stochastic properties.

In Section 2, we start with a detailed discussion about ordinal distances and their properties. Afterward in Section 3, we define distance-based analytic tools for (marginal) properties of ordinal random variables like location, dispersion, or skewness. The corresponding sample measures, as computed from an iid ordinal sample, are investigated in Section 4, and we apply them to a survey dataset about the economic situation in Germany in Section 5. Distance-based measures can also be used for quantifying the dependence of ordinal random variables, see

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Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/JASA.

Section 6. Section 7 then moves on to the time series case, where marginal properties of ordinal time series as well as a novel class of measures of serial dependence are considered. For illustration, we consider a set of time series of monthly credit ratings of countries in the European Union in Section 8. Finally, Section 9 concludes.

2. Ordinal Distances

Let X be a random variable with some range $\mathcal{S}$ (state space). Although the main focus of this article is on ordinal random variables, that is, having the ordered categorical range $\mathcal{S} = \{s_0, \dots, s_m\}$ with $s_0 < \dots < s_m$, we shall sometimes also consider different types of range for illustrative purposes, like $\mathbb{R}$ being the set of real numbers. We start with the general definition of distance measures.

Definition 2.1 (Distance measures). The mapping $d : \mathcal{S} \times \mathcal{S} \to [0; \infty)$ is said to be a *distance measure* on $\mathcal{S}$ if it satisfies for all $x, y \in \mathcal{S}$:

(D1) If $x = y$, then $d(x, y) = 0$. (Positive semidefinite)
(D2) $d(x, y) = d(y, x)$. (Symmetry)

The distance measure d is said to be a *pseudo-metric* if it even satisfies

(D3) $d(x, z) \le d(x, y) + d(y, z)$ for all $x, y, z \in \mathcal{S}$. (Triangular inequality)

The pseudo-metric d is said to be a *metric* if it even satisfies

(D1') $x = y$ iff $d(x, y) = 0$ for all $x, y \in \mathcal{S}$. (Positive definite)

If the range $\mathcal{S}$ is ordinal, further requirements on the distance could be relevant, for example, to guarantee that the distance is compatible with the ordering.

Definition 2.2 (Possible properties of ordinal distances). Let $d : \mathcal{S} \times \mathcal{S} \to [0; \infty)$ be a distance measure (Definition 2.1) defined on the ordinal range $\mathcal{S} = \{s_0, \dots, s_m\}$ with $s_0 < \dots < s_m$. It may satisfy some of the following properties:

(O1) $d(s_0, s_m) = \max_{x,y \in \mathcal{S}} d(x, y)$. (Maximization)
(O2) d is said to be compatible with the ordering if

$$x < y < z \quad \text{implies that} \quad d(x, z) > d(x, y),\ d(y, z).$$

(O3) d is said to be additive if for given $d_1, \dots, d_m > 0$, it holds that

$$d(s_i, s_{i+k}) = d_{i+1} + \dots + d_{i+k} \quad \text{for all } i = 0, \dots, m-1;\ k = 1, \dots, m-i.$$

(O4) $d(s_i, s_j) = d(s_{m-i}, s_{m-j})$ for all $0 \le i < j \le m$. (Centrosymmetry)

Note that in Definition 2.2, we have the implications (O3) $\Rightarrow$ (O2) $\Rightarrow$ (O1), so the maximal pairwise distance is equal to $d(s_0, s_m)$ if d is compatible with the ordering or even additive. Property (O4) in Definition 2.2 will be relevant in Section 3.3, where we discuss the possible symmetry of an ordinal distribution.

The most simple type of distance measure is the *Hamming distance*, which checks its arguments for equality. If applied to ordinal data with the ordered categorical range $\mathcal{S}$, it is given by

$$d_{\mathrm{H}}(s_k, s_l) := 1 - \delta_{k,l}, \tag{1}$$

where $\delta_{k,l}$ denotes the Kronecker delta. However, d_{H} does not make use of the ordering of categories (actually, it can be applied to purely nominal data as well). In particular, it does not satisfy property (O2) of Definition 2.2 (but properties (O1) and (O4), and it is also a metric). Ordinal distances satisfying (O2) can be constructed by applying types of Minkowski distance to the rank counts. Applying the *block distance*, we obtain

$$d_{\mathrm{o},1}(s_k, s_l) := |k - l|, \tag{2}$$

which expresses the number of categories between s_k and s_l. So $d_{\mathrm{o},1}$ treats the ordinal data as if assigning equidistant numbers (scores) to the categories, see Labovitz (1970) and the subsequent discussion of that article. As a consequence, it satisfies all properties in Definition 2.2. $d_{\mathrm{o},1}$ is easily adapted to cases where successive categories are equidistant but have pairwise distances not normalized to one.

The *squared Euclidean distance*, in contrast, causes a quadratic penalty

$$d_{\mathrm{o},2}(s_k, s_l) := (k - l)^2. \tag{3}$$

Generally, $d_{\mathrm{o},2}$ does not satisfy the triangular inequality (D3) in Definition 2.1 and is thus not a metric (but a positive definite distance). Concerning Definition 2.2, it does not satisfy property (O3) but all remaining properties given there.

Note that the distance measures (1)–(3) are defined such that the distance values do not depend on the actual labeling $s_0, \dots, s_m$ of the ordinal states. The labels of an ordinal scale could generally be modified by any strictly monotonic increasing mapping (Stevens 1959, Table 1), but these would not affect the above distance values. As a result, when we later define ordinal statistics based on expected distances, these statistics will be invariant under the permitted scale transformations, which is a criterion for the appropriateness of a statistic (Stevens 1959, pp. 27–28).

The appropriate choice of weights for the dissimilarities within an ordinal range depends on the application context, and it can be expressed by an appropriate distance measure d on $\mathcal{S}$. In contrast to the basic distance measures (2) and (3), successive ordinal states are not necessarily "equally spaced" but may have different distances, also see Section 8. As an example, consider the grading of an exam. At the author's university, the grades vary between 1 (best) and 5 (worst), where the grades $1, 1.3, \dots, 3.7, 4$ are considered a pass, and $4.3, 5$ a fail. In this example, the distance between 4 and 4.3 might be judged larger than the remaining successive distances. Another example is reported by Taleb and Limam (2002), which is about the quality monitoring of manufactured porcelain products. These are classified as either "standard" (S), "second choice" (SC), "third choice" (TC), or "chipped" (C), with C being probably more distant from TC than TC from SC; also see the related discussion in Tucker, Woodall, and Tsui (2002). For this reason, it is important to define analytic approaches for ordinal data that allow for *any* kind of distance measure, beyond the simple examples of Hamming and Minkowski distances, although we shall further use the latter for illustrative purposes.

3. Distance-Based Analysis of Ordinal Random Variables

It is proposed to quantify properties of an ordinal random variable, such as location, dispersion, or skewness, by measures based on *expected distances*. Note that since the distance between two ordinal random variables is a nonnegative real number according to Definition 2.1 (an analogous statement holds for other types of random variable as well), the expectation thereof is well defined. It shall turn out that this distance-based approach, using appropriate types of distance function, covers measures of marginal properties already existing in the literature, also see the overview table in Supplement D. But since any other distance function defined on $\mathcal{S}$ can be used as well, this approach offers the potential for obtaining further tailor-made measures of marginal properties.

In the sequel, we use the following notations: If X is a categorical random variable with range $\mathcal{S} = \{s_0, \ldots, s_m\}$, then the vector of marginal probabilities is denoted by $\boldsymbol{p} = (p_0, \ldots, p_m)^\top$, where $p_i = P(X = s_i)$. So $\boldsymbol{p}$ summarizes the probability mass function (PMF). If $\mathcal{S}$ is even ordinal, then we denote the cumulative probabilities by $f_k = \sum_{l=0}^{k} p_l = P(X \leq s_k)$ and $\boldsymbol{f} = (f_0, \ldots, f_{m-1})^\top$.

3.1. Location

Expected distances can be used to express the location of a random variable X. Location can be defined as a minimizer of expected distance, that is,

$$x_{\mathrm{loc},d} := \arg\min_{x\in\mathcal{S}} E\big[d(X,x)\big], \tag{4}$$

see Marden (1995). Then $x_{\mathrm{loc},d} \in \mathcal{S}$ holds although $x_{\mathrm{loc},d}$ is not necessarily uniquely determined. In the case of a one-point distribution (i.e., where all probability mass concentrates on one category), $x_{\mathrm{loc},d}$ is equal to this category, as required for a location measure by Bickel and Lehmann (1975). As an example, it is well known that the Hamming distance is minimized by the mode(s), and the block distance by the median. If using the squared Euclidean distance (3), then $E\big[d_{\mathrm{o},2}(X, s_i)\big] = E\big[(I - i)^2\big] = V[I] + \big(E[I] - i\big)^2$. Minimizing this distance for $i \in \{0, \ldots, m\}$, we end up with the category corresponding to the rounded value of $E[I]$, that is, with $s_{\mathrm{round}(E[I])}$. Note that $E[I] = m - \sum_{i=0}^{m-1} f_i$, see Supplement A.1.

Alternatively, we may define location with respect to the lowest category s_0 by considering the expected distance from s_0. Then one might choose that $s \in \mathcal{S}$ the distance $d(s, s_0)$ of which is closest to $E\big[d(X, s_0)\big]$. If using the ordinal block distance (2) as the distance measure, for example, then $E\big[d_{\mathrm{o},1}(X, s_0)\big] = E[I]$ and the location would thus be given by $s_{\mathrm{round}(E[I])}$.

3.2. Dispersion

If location is measured as $x_{\mathrm{loc},d} := \arg\min_x E\big[d(X,x)\big]$, then a possible measure of dispersion (spread) is $\mathrm{disp}_{\mathrm{loc},d}(X) := E\big[d(X, x_{\mathrm{loc},d})\big]$, see Marden (1995). One may also normalize this kind of spread by dividing $\mathrm{disp}_{\mathrm{loc},d}(X)$ by $\max_{x,y\in\mathcal{S}} d(x,y)$, where the latter equals $d(s_0, s_m)$ if property (O1) holds.

An alternative approach for measuring the dispersion (variability) of X is to use the expected distance between independent copies X_1, X_2 of X, that is, $E\big[d(X_1, X_2)\big]$. This approach has been referred to as the *diversity coefficient* (*DIVC*) by Rao (1982), and it was also considered by Gadrich, Bashkansky, and Zitikis (2015). In the ordinal case, the DIVC becomes

$$\mathrm{disp}_d(X) := E\big[d(X_1,X_2)\big] = \textstyle\sum_{i,j=0}^{m} d(s_i,s_j)\, p_i\, p_j = \boldsymbol{p}^\top \mathbf{D}\, \boldsymbol{p}, \tag{5}$$

where $\mathbf{D} = \big(d(s_i,s_j)\big)_{i,j=0,\ldots,m}$ denotes the distance matrix. Obviously, if d is positive definite, then the lower bound 0 is only reached if X follows a one-point distribution. Furthermore, disp_d is bounded from above by $d(s_0, s_m)$ if property (O1) holds. The precise upper bound is derived in Supplement A.2 using differential calculus. For disp_d as a function of $p_0, \ldots, p_{m-1}$ (with $p_m = 1 - p_0 - \cdots - p_{m-1}$), it is first shown that the partial derivatives $\frac{\partial}{\partial p_k}\mathrm{disp}_d(X)$ equal $2\big(E\big[d(X, s_k)\big] - E\big[d(X, s_m)\big]\big)$. Then the maximum is derived depending on the considered distance measure.

If X is a categorical random variable with range $\mathcal{S} = \{s_0, \ldots, s_m\}$ and marginal probabilities $\boldsymbol{p}$, and if d is chosen as the Hamming distance (1), then

$$\mathrm{disp}_{d_{\mathrm{H}}}(X) = E\big[1 - \delta_{X_1,X_2}\big] = 1 - \textstyle\sum_{i=0}^{m} p_i^2 =: 1 - s_2(\boldsymbol{p}), \tag{6}$$

which is equal to the *index of qualitative variation* (also *Gini index*) $\mathrm{IQV} = \frac{m+1}{m}\cdot\big(1 - s_2(\boldsymbol{p})\big)$ except the normalizing factor $\frac{m+1}{m}$, see Rao (1982) and Kvålseth (1995) as well as Supplement A.2. $\mathrm{disp}_{d_{\mathrm{H}}}$ is minimized by a one-point distribution $\boldsymbol{p}_{\mathrm{one}}$, and maximized by the uniform distribution $\boldsymbol{p}_{\mathrm{uni}}$ on $\mathcal{S}$. Thus, $\mathrm{disp}_{d_{\mathrm{H}}}$ expresses the uncertainty in forecasting the outcome of X, which is meaningful not only in the nominal but also in the ordinal case.

It is, however, more common to consider the extreme two-point distribution $\boldsymbol{p}_{\mathrm{two}} = (0.5, 0, \ldots, 0, 0.5)^\top$ (polarized distribution, i.e., with $P(X = s_0) = P(X = s_m) = 0.5$) as the case of maximal ordinal dispersion, as it expresses maximal dissent, whereas the one-point distribution goes along with maximal consensus (Kiesl 2003). A corresponding dispersion measure is obtained if using (5) together with the block distance. For a real-valued random variable X (with existing moments), the DIVC together with the block distance leads to *Gini's mean difference*, $E\big[|X_1 - X_2|\big]$, a well-known alternative to the variance with closed-form expressions for a number of parametric distributions (discrete and continuous), see Ramasubban (1958), Yitzhaki (2003), and the references therein. If applying the ordinal block distance (2), we obtain

$$\mathrm{disp}_{d_{\mathrm{o},1}} = \textstyle\sum_{k,l=0}^{m} |k-l|\, p_k p_l = 2\sum_{k=0}^{m-1} f_k(1 - f_k), \tag{7}$$

where the latter equality, relying on the cumulative probabilities f_k, was shown by Blair and Lacy (1996). As for any additive distance, $\mathrm{disp}_{d_{\mathrm{o},1}}$ has to be normalized by the factor $2/d_{\mathrm{o},1}(s_0, s_m) = 2/m$, see Theorem A.2.2 in the supplement. Then (7) leads to the *index of ordinal variation* (Kvålseth 1995)

$$\mathrm{IOV} = \tfrac{4}{m}\textstyle\sum_{k=0}^{m-1} f_k(1 - f_k) = 1 - \tfrac{1}{m}\sum_{k=0}^{m-1}(2f_k - 1)^2.$$

It should be noted that also the variance of a real-valued random variable X is included in the DIVC approach (5) by choosing d as the squared Euclidean distance (Cuadras, Fortiana, and Oliva 1997): $E\big[(X_1 - X_2)^2\big] = 2\,V[X]$. The ordinal counterpart, obtained by using (3) within (5), is given by

$$\mathrm{disp}_{d_{o,2}} = E\big[(I_1 - I_2)^2\big] = 2\,V[I], \tag{8}$$

where $V[I] = \sum_{i=0}^{m-1} f_i\,\big(1 - \sum_{j=0}^{m-1} f_j\big) + 2\,\sum_{i=1}^{m-1}\sum_{j=0}^{i-1} f_j$, see Supplement A.1. The normalized version is given by $\frac{2}{m^2}\,\mathrm{disp}_{d_{o,2}} = \frac{4}{m^2}\,V[I]$, see Supplement A.2. Weisberg (1992) states on p. 67 of his book that "the variance and standard deviation are frequently employed even on ordinal data"; in view of the presented distance approach, this practice appears to be justified if quadratic penalties apply to rank count differences.

Supplement B provides some numerical comparisons regarding the dispersion measures $\mathrm{disp}_{d_{\mathrm{H}}}, \mathrm{disp}_{d_{o,1}}, \mathrm{disp}_{d_{o,2}}$ for different types of ordinal distributions. Among others, it is illustrated that $\mathrm{disp}_{d_{\mathrm{H}}}$ might lead to quite different results than $\mathrm{disp}_{d_{o,1}}, \mathrm{disp}_{d_{o,2}}$ because of a different understanding of maximal dispersion.

3.3. Symmetry

If X is an ordinal random variable with $P(X = s_i) = p_i$, then we call X^{r} a reflected copy of X if $P(X = s_i) = p_{m-i}$. Equivalently, if I is the rank count variable corresponding to X, then $X^{\mathrm{r}} = s_{m-I}$. It should be noted that the DIVC of X^{r} according to (5) computes as $\mathrm{disp}_d(X^{\mathrm{r}}) = \sum_{i,j=0}^{m} d(s_{m-i}, s_{m-j})\,p_i\,p_j$, and this equals $\mathrm{disp}_d(X)$ if the distance measure d is centrosymmetric in the sense of property (O4) in Definition 2.2. So if considering the symmetry of an ordinal distribution, it appears reasonable to require d to be centrosymmetric such that the dispersion of X does not change by simply reflecting the random variable. Then, a distribution is said to be *symmetric* if $p_i = p_{m-i}$ for all $i = 0, \ldots, m$, that is, if X and X^{r} are identically distributed. Equivalently, $f_i = 1 - f_{m-1-i}$ has to hold for $i = 0, \ldots, m-1$, see Klein and Doll (2018).

Expected distances can now be used for quantifying the extent of asymmetry of the ordinal distribution. Let X^{r} denote an independent and reflected copy of the ordinal random variable X, that is, let $X = s_{I_1}$ and $X^{\mathrm{r}} = s_{m-I_2}$ with iid I_1, I_2. Then we can check for symmetry by comparing X with X^{r} in terms of $E\big[d(X, X^{\mathrm{r}})\big]$. Obviously, this expected distance equals the DIVC $\mathrm{disp}_d(X)$ from (5) if the distribution of X is symmetric. Therefore, we measure the extent of *asymmetry* of X by

$$\begin{aligned}\mathrm{asym}_d(X) &= E\big[d(X, X^{\mathrm{r}})\big] - \mathrm{disp}_d(X)\\ &= \sum_{i,j=0}^{m} d(s_i, s_j)\,p_i(p_{m-j} - p_j).\end{aligned} \tag{9}$$

If the maximization property (O1) holds, then $E\big[d(X, X^{\mathrm{r}})\big] \le d(s_0, s_m)$ whereas $\mathrm{disp}_d(X) \ge 0$. Therefore, $\mathrm{asym}_d(X) \le d(s_0, s_m)$ and this upper bound is reached for the extreme one-point distributions $p_0 = 1$ or $p_m = 1$ (maximal asymmetry). But depending on the actual distance measure, the upper bound might also be reached for other distributions. Such an example is the Hamming distance with $\mathrm{asym}_{d_{\mathrm{H}}} = \sum_{i=0}^{m} p_i\,(p_i - p_{m-i})$, as shown in Example A.3.1 in the supplement.

If the distribution of X is symmetric, then $\mathrm{asym}_d = 0$, which, unfortunately, is not necessarily the minimal value of asym_d: depending on the actual distance measure, there might be asymmetric distributions leading to an even smaller value than in the symmetry case. A sufficient condition to guarantee that $\mathrm{asym}_d \ge 0$ is derived in Supplement A.3. Let $\mathbf{J} = (\delta_{i,m-j})_{i,j=0,\ldots,m}$ denote the $(m+1)$-dimensional exchange matrix (counteridentity matrix), and let $\mathbf{I}$ be the $(m+1)$-dimensional identity matrix. Then, using that $\mathbf{D}$ is centrosymmetric, one can rewrite asym_d as the quadratic form $\boldsymbol{p}^\top\,(\mathbf{J} - \mathbf{I})\mathbf{D}\,\boldsymbol{p}$. So $\mathrm{asym}_d \ge 0$ if $(\mathbf{J} - \mathbf{I})\mathbf{D}$ is positive semidefinite. In this case, we can normalize asym_d to the interval $[0; 1]$ by dividing by $d(s_0, s_m)$.

None of the aforementioned problems happens if using the squared Euclidean distance (3), then

$$\mathrm{asym}_{d_{o,2}} \overset{(8)}{=} E\big[(I_1 + I_2 - m)^2\big] - 2\,V[I] = m^2\,\big(1 - \tfrac{2}{m}\,E[I]\big)^2, \tag{10}$$

which is minimized exactly for $E[I] = \frac{m}{2}$ (this covers the symmetry case), and maximized exactly for $E[I] \in \{0, m\}$ (maximal asymmetry). For the normalized measure, it holds that $\frac{1}{m^2}\,\mathrm{asym}_{d_{o,2}} \le 1 - \frac{2}{m^2}\,\mathrm{disp}_{d_{o,2}}$.

Analogous conclusions hold if using the ordinal block distance (2), see Supplement A.3, then

$$\mathrm{asym}_{d_{o,1}} = \textstyle\sum_{j=0}^{m-1}(1 - f_j - f_{m-j-1})^2. \tag{11}$$

$\mathrm{asym}_{d_{o,1}}$ takes its minimal value 0 *iff* all $f_j = 1 - f_{m-j-1}$ (symmetry). Furthermore, we have $-1 \le 1 - f_j - f_{m-j-1} \le 1$, so $\mathrm{asym}_{d_{o,1}}$ reaches its maximal value m iff all summands $(1 - f_j - f_{m-j-1})^2 = 1$. This happens iff either $f_0 = 1$ (and thus $f_j = 1$ for all j) or $f_0 = \cdots = f_{m-1} = 0$, that is, in the extreme asymmetry cases. Also note that the normalized version of $\mathrm{asym}_{d_{o,1}}$ satisfies $\frac{1}{m}\,\mathrm{asym}_{d_{o,1}} \le 1 - \mathrm{IOV}$.

There are different ways of defining the *skewness* of an ordinal random variable, see Klein and Doll (2018), but maximal positive (negative) skewness is uniquely characterized by requiring $p_0 = 1$ ($p_m = 1$), that is, the extreme left (right) one-point distribution. A signed skewness measure is obtained by the approach

$$\mathrm{skew}_d(X) = E\big[d(X, s_m)\big] - E\big[d(X, s_0)\big]. \tag{12}$$

If d is centrosymmetric (O4), then $\mathrm{skew}_d = 0$ if the distribution is symmetric. If the distance d satisfies the triangular inequality (D3) or the maximization property (O1), then skew_d takes the maximal (minimal) value $d(s_0, s_m)$ ($-d(s_0, s_m)$) for $p_0 = 1$ ($p_m = 1$), see Supplement A.3. So we can normalize skew_d to the interval $[-1; 1]$ by dividing by $d(s_0, s_m)$.

In case of the Hamming distance, we obtain $\mathrm{skew}_{d_{\mathrm{H}}} = p_0 - p_m$ being a rather crude skewness measure. If using the ordinal block distance (2) for skew_d (together with the normalizing factor $\frac{1}{m}$), then it follows that

$$\tfrac{1}{m}\,\mathrm{skew}_{d_{o,1}} = 1 - \tfrac{2}{m}\,E[I] = \tfrac{2}{m}\,\textstyle\sum_{i=0}^{m-1} f_i - 1, \tag{13}$$

see Supplement A.3. $\frac{1}{m}\,\mathrm{skew}_{d_{o,1}}$ equals 0 for a symmetric distribution (more generally, for an ordinal distribution with $E[I] = \frac{m}{2}$), and it equals 1 (-1) for maximal positive (negative) skewness. This measure coincides with the measure $S^{\mathrm{K}} = S^{b_1}$ in Klein and Doll (2018). Also note that $\mathrm{asym}_{d_{o,2}}$ from (10) just equals $\mathrm{skew}^2_{d_{o,1}}$. Finally, if using the squared Euclidean distance (3), then again

$$\tfrac{1}{m^2}\,\mathrm{skew}_{d_{o,2}} = \tfrac{1}{m^2}\,\Big(E\big[(I - m)^2\big] - E\big[(I - 0)^2\big]\Big) = 1 - \tfrac{2}{m}\,E[I].$$

So any of the measures $\mathrm{asym}_{d_{o,2}}, \mathrm{skew}_{d_{o,1}}, \mathrm{skew}_{d_{o,2}}$ evaluates symmetry by checking if the rank count's mean $E[I]$ is halfway between 0 and m.

The numerical illustrations in Supplement B show the behavior of the asymmetry and skewness measures relying on either d_{H} or $d_{o,1}$ for different types of asymmetry. It is also shown that $\mathrm{skew}_{d_{o,1}}$ has to be interpreted with some caution, as it might take the value 0 also for some asymmetric distributions.

4. Distance-Based Analysis of iid Ordinal Samples

While in Section 3, we proposed measures of location, dispersion, and symmetry for ordinal random variables, we now consider corresponding sample counterparts to be computed from iid replications $X_1, \ldots, X_n$ of the ordinal random variable X. Here, we exclude the case of X following a one-point distribution ($\mathrm{disp}_d(X) = 0$) and thus being deterministic. We denote the sample counterparts of $p_i, \boldsymbol{p}, f_k, \boldsymbol{f}$ by $\hat{p}_i, \hat{\boldsymbol{p}}, \hat{f}_k, \hat{\boldsymbol{f}}$, and these are computed as the respective relative frequencies from the given sample data $x_1, \ldots, x_n$.

The basic idea for defining sample versions of the ordinal measures is to replace the expected distances by corresponding sample means of distance values. For the location measure $x_{\mathrm{loc},d}$ according to (4) in Section 3.1, which is defined as that category $x \in \mathcal{S}$ that minimizes $E\big[d(X,x)\big]$, we obtain the sample version as $\hat{x}_{\mathrm{loc},d} := \arg\min_x \frac{1}{n} \sum_{t=1}^{n} d(x_t, x)$.

One of the dispersion measures discussed in Section 3.2 was related to $x_{\mathrm{loc},d}$, namely $\mathrm{disp}_{\mathrm{loc},d} := E\big[d(X, x_{\mathrm{loc},d})\big]$; an obvious sample version is given by $\frac{1}{n} \sum_{t=1}^{n} d(x_t, \hat{x}_{\mathrm{loc},d})$. If using the DIVC approach $E\big[d(X_1, X_2)\big]$ for quantifying dispersion instead, then an unbiased estimator (for iid data) is given by $\frac{2}{n(n-1)} \sum_{i<j} d(x_i, x_j)$, also see the works by Nair (1936) and Lomnicki (1952). So the general approach for estimating the ordinal DIVC (5) is given by $\frac{n}{n-1}\, \widehat{\mathrm{disp}}_d$ with

$$\widehat{\mathrm{disp}}_d \;=\; \sum_{i,j=0}^{m} d(s_i, s_j)\, \hat{p}_i\, \hat{p}_j \;=\; \hat{\boldsymbol{p}}^\top \mathbf{D}\, \hat{\boldsymbol{p}}. \tag{14}$$

Analogously, sample versions of asym_d and skew_d are defined by

$$\widehat{\mathrm{asym}}_d = \hat{\boldsymbol{p}}^\top (\mathbf{J} - \mathbf{I})\mathbf{D}\, \hat{\boldsymbol{p}}, \quad \widehat{\mathrm{skew}}_d = \sum_{i=0}^{m} \Big(d(s_i, s_m) - d(s_i, s_0) \Big)\, \hat{p}_i. \tag{15}$$

For special cases of the distance measure d, however, it might be more convenient to use a tailor-made sample version. As an example, an alternative sample version of $\mathrm{disp}_{d_{o,1}}$ from (7) is obtained by replacing the f_k by $\hat{f}_k$, that is, the relative cumulative frequencies computed from $x_1, \ldots, x_n$. Sample versions of the particular asymmetry measure $\mathrm{asym}_{d_{o,1}}$ from (11) and the skewness measure $\mathrm{skew}_{d_{o,1}}$ from (13) are obtained analogously.

To compute confidence intervals for the proposed measures, or to do hypothesis tests (e.g., with regard to the null hypothesis of a symmetric distribution), the (asymptotic) distribution of the sample measures is required. Here as well as later in Section 7, the starting point is to establish the asymptotic normality of $\sqrt{n}\,\hat{\boldsymbol{p}}$ and $\sqrt{n}\,\hat{\boldsymbol{f}}$, see Kiesl (2003) and Weiß (2013) as well as Supplement A.5. Then we use a linear expansion of the considered statistic for establishing its asymptotic normality (Delta method), and a quadratic expansion for correcting the bias (if necessary at all), see Supplements A.6 and A.7. Unfortunately, it is not possible to find simple closed-form formulas for these asymptotics being valid for any type of distance measure d. But focusing on a particular type of distance instead, such derivations are possible. In what follows, we consider the case of the ordinal block distance $d_{o,1}$ for illustration, as this is probably the most often used distance measure for ordinal data. The resulting dispersion and symmetry measures rely on the CDF of the ordinal random variables, and such measures have also been considered by Kiesl (2003), Kvålseth (2011a), and Klein and Doll (2018) for ordinal data. The sample CDF is shown to be asymptotically normally distributed, $\sqrt{n}\,(\hat{\boldsymbol{f}} - \boldsymbol{f}) \xrightarrow{\mathrm{d}} \mathrm{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\Sigma} = (\sigma_{i,j})_{i,j=0,\ldots,m-1} \text{ given by } \sigma_{i,j} \;=\; f_{\min\{i,j\}} - f_i f_j, \tag{16}$$

see Kiesl (2003) as well as Supplement A.5. The derived stochastic properties (see Supplement A.6) of the sample measures of dispersion and symmetry are summarized in the following theorem.

Theorem 4.1. Let $X_1, \ldots, X_n$ be iid ordinal random variables with range $\mathcal{S} = \{s_0, \ldots, s_m\}$ and the corresponding distance measure $d_{o,1}$.

(i) If $(f_0, \ldots, f_{m-1}) \neq (\frac{1}{2}, \ldots, \frac{1}{2})$ (extreme two-point distribution), then $\widehat{\mathrm{disp}}_{d_{o,1}}$ is asymptotically normally distributed with exact mean $(1 - \frac{1}{n})\, \mathrm{disp}_{d_{o,1}}$ and approximate variance

$$\tfrac{4}{n} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j)\, \big(f_{\min\{i,j\}} - f_i f_j\big).$$

(ii) If the distribution of the X_i is not symmetric, then $\widehat{\mathrm{asym}}_{d_{o,1}}$ is asymptotically normally distributed with exact mean

$$\mathrm{asym}_{d_{o,1}} + \tfrac{1}{n}\, \mathrm{disp}_{d_{o,1}} + \tfrac{2}{n} \sum_{j=0}^{m-1} \big(f_{\min\{j,m-j-1\}} - f_j f_{m-j-1}\big)$$

and approximate variance

$$\tfrac{16}{n} \sum_{i,j=0}^{m-1} (1 - f_i - f_{m-i-1})(1 - f_j - f_{m-j-1})\, \big(f_{\min\{i,j\}} - f_i f_j\big).$$

(iii) $\widehat{\mathrm{skew}}_{d_{o,1}}$ is asymptotically normally distributed with exact mean $\mathrm{skew}_{d_{o,1}}$ and variance

$$\tfrac{4}{n} \sum_{i,j=0}^{m-1} \big(f_{\min\{i,j\}} - f_i f_j\big).$$

Parts (i)–(iii) are easily adapted to the respective normalized versions $\widehat{\mathrm{IOV}} = \frac{2}{m}\, \widehat{\mathrm{disp}}_{d_{o,1}}, \frac{1}{m}\, \widehat{\mathrm{asym}}_{d_{o,1}}$ as well as $\frac{1}{m}\, \widehat{\mathrm{skew}}_{d_{o,1}}$.

Note that part (i) of Theorem 4.1 is already known from Kiesl (2003) and part (iii) from Klein and Doll (2018).

Remark 4.2. As argued in Supplement A.6, the asymptotic variance in part (i) of Theorem 4.1 is not valid in the extreme two-point case (actually, the given expression then reduces to 0), because there does not exist a linear approximation of $\widehat{\mathrm{disp}}_{d_{o,1}}$ (but the bias formula still holds true). The same arguments apply to $\widehat{\mathrm{asym}}_{d_{o,1}}$ if the distribution of the X_i is symmetric, that is, if $f_{m-1-i} = 1 - f_i$ holds for all $i = 0, \ldots, m-1$. These boundary cases, however, might be particularly interesting for testing purposes, for example, to establish a significant asymmetry. As shown in Supplement A.6, the following results hold:

(i) If $(f_0,\ldots,f_{m-1}) = (\frac{1}{2},\ldots,\frac{1}{2})$ (extreme two-point distribution), then $\widehat{\mathrm{IOV}} = \frac{2}{m}\,\widehat{\mathrm{disp}}_{d_{o,1}}$ is asymptotically distributed as $1-\frac{1}{n}Z^2$ with Z being an $\mathrm{N}(0,1)$-variate. In particular, $\widehat{\mathrm{disp}}_{d_{o,1}}$ has the mean $\frac{m}{2}\,(1-\frac{1}{n})$ and approximate variance $\frac{m^2}{2}\,\frac{1}{n^2}$.

(ii) If the distribution of the X_i is symmetric, then $n\,\widehat{\mathrm{asym}}_{d_{o,1}}$ is asymptotically distributed like $2\sum_{i=1}^{u}\lambda_i Z_i^2$ with $Z_1,\ldots,Z_u$ being iid $\mathrm{N}(0,1)$-variates, where $\lambda_1,\ldots,\lambda_u$ denote the nonzero eigenvalues of $\mathbf{\Sigma}\,(\mathbf{I}+\mathbf{J})$. Here, $\mathbf{\Sigma}$ is given by (16), and $\mathbf{I}$ ($\mathbf{J}$) denotes the m-dimensional identity matrix (exchange matrix). As a consequence,

$$E\big[\widehat{\mathrm{asym}}_{d_{o,1}}\big] \overset{a}{=} \frac{2}{n}\sum_{i=1}^{u}\lambda_i, \qquad V\big[\widehat{\mathrm{asym}}_{d_{o,1}}\big] \overset{a}{=} \frac{8}{n^2}\sum_{i=1}^{u}\lambda_i^2.$$

The finite-sample performance of the asymptotic approximations implied by Theorem 4.1 and Remark 4.2 is investigated in Supplement C.1, where results from a simulation study are presented.

Table 1. Normalized sample measures $\frac{2}{m}\,\widehat{\mathrm{disp}}_{d_{o,1}}$, $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$, and $\frac{1}{m}\,\widehat{\mathrm{skew}}_{d_{o,1}}$ (point estimates and 95 % confidence intervals) computed for questions 1–4 of "ALLBUS 2016" data, see Section 5.

$\frac{2}{m}\,\widehat{\mathrm{disp}}_{d_{o,1}}$:	
Q1: 0.384 (0.372; 0.396)	Q3: 0.326 (0.316; 0.337)
Q2: 0.385 (0.371; 0.400)	Q4: 0.271 (0.259; 0.284)
$\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$:	
Q1: 0.162 (0.150; 0.174)	Q3: 0.027 (0.026; 0.032)
Q2: 0.179 (0.166; 0.192)	Q4: 0.009 (0.006; 0.011)
$\frac{1}{m}\,\widehat{\mathrm{skew}}_{d_{o,1}}$:	
Q1: −0.324 (−0.337; −0.312)	Q3: 0.128 (0.117; 0.139)
Q2: −0.329 (−0.342; −0.316)	Q4: −0.071 (−0.081; −0.061)

5. Application: Economic Situation

The German General Social Survey ("ALLBUS") is carried out by the "GESIS—Leibniz Institute for the Social Sciences" every second year since 1980. We analyze a part of the data collected during the most recent survey in 2016, see GESIS (2017) for the full dataset. While Klein and Doll (2018) discussed a question about the political "left-right self-reporting" from this survey, we consider the first four questions about the economic situation in Germany:

- Q1: How would you generally rate the current economic situation in Germany?
  Q2: And your own current financial situation?
  Possible answers:
  s_0 (very bad), s_1 (bad), s_2 (partly good/partly bad), s_3 (good), s_4 (very good).
- Q3: What do you think the economic situation in Germany will be like in one year?
  Q4: And what will your own financial situation be like in one year?
  Possible answers:
  s_0 (considerably worse than today), s_1 (somewhat worse than today), s_2 (the same), s_3 (somewhat better than today), s_4 (considerably better than today).

We treat successive ordinal categories as being equidistant (w. l. o. g., this distance is normalized to 1), so the distance measure $d_{o,1}$ is relevant here. The numbers of valid answers per question are $n_1 = 3482$, $n_2 = 3489$, $n_3 = 3459$, and $n_4 = 3464$. The corresponding sample CDF vectors are $\hat{\boldsymbol{f}}_1 = (0.007, 0.061, 0.376, 0.907)^\top$, $\hat{\boldsymbol{f}}_2 = (0.017, 0.087, 0.318, 0.921)^\top$, $\hat{\boldsymbol{f}}_3 = (0.025, 0.322, 0.910, 0.998)^\top$, and $\hat{\boldsymbol{f}}_4 = (0.009, 0.082, 0.786, 0.981)^\top$, so the median answer to Q1 & Q2 (current situation) is s_3 (good), and the median answer to Q3 & Q4 (situation in one year) is s_2 (the same).

Let us now have a look at further characteristics of these ordinal distributions, see Table 1 for a summary (always rounded to three decimal places). The given 95 % confidence intervals are computed according to the normal approximation of Theorem 4.1. Furthermore, since the sample sizes are very large, a bias correction according to Theorem 4.1 does not have an effect on the displayed decimal places of the given point estimates.

The distributions $\hat{\boldsymbol{f}}_1$ and $\hat{\boldsymbol{f}}_2$ do not only have the same median, they also show a nearly identical dispersion value and a very similar symmetry behavior: both are clearly asymmetric in form of negative skewness, and the null hypothesis of a symmetric distribution is rejected by both tests. So the current economic situation is rated by the respondents nearly the same for whole Germany and for themselves personally.

Things change if we look at the distributions $\hat{\boldsymbol{f}}_3$ and $\hat{\boldsymbol{f}}_4$ referring to the economic situation one year later. Although the medians are again the same, the dispersion values are now separated from each other, in the sense that the confidence intervals do not overlap. Actually, the dispersion regarding the own future economic situation is less than for the German situation (and both are notably lower than for $\hat{\boldsymbol{f}}_1,\hat{\boldsymbol{f}}_2$). Even more pronounced differences are observed with regard to the symmetry behavior of $\hat{\boldsymbol{f}}_3,\hat{\boldsymbol{f}}_4$. Although both symmetry tests again reject the null of symmetry, the extend of asymmetry is much lower than before, and while $\hat{\boldsymbol{f}}_4$ is again negatively skewed, $\hat{\boldsymbol{f}}_3$ shows positive skewness. So while the median rating is s_2 (the same) for both Q3 & Q4, the difference in skewness implies that the future economic situation is actually rated worse for whole Germany than for the respondents personally.

6. On the Dependence of Ordinal Random Variables

As an intermediate step before turning to ordinal time series in Section 7, let us briefly consider the situation of a pair (X, Y) of ordinal random variables defined on the same range $\mathcal{S}$. We denote the (cumulative) probabilities and rank counts of X, Y as before, but with the additional subscript X or Y, respectively: $p_{X;i}, p_{Y;j}, f_{X;i}, f_{Y;j}$ and I_X, I_Y. In addition, we introduce the following notations for the joint (cumulative) probabilities: $p_{XY;ij} = P(X = s_i, Y = s_j)$ and $f_{XY;ij} = P(X \leq s_i, Y \leq s_j)$. Then X and Y are independent of each other if the joint probabilities factorize as $p_{XY;ij} = p_{X;i}\, p_{Y;j}$ and $f_{XY;ij} = f_{X;i}\, f_{Y;j}$ for all $i, j = 0, \ldots, m$.

Inspired by the DIVC (5), a distance-based approach for measuring the dependence between X and Y might be based upon

$$E\big[d(X,Y)\big] = \textstyle\sum_{i,j=0}^{m} d(s_i, s_j)\, p_{XY;ij}, \tag{17}$$

which simplifies to $E_{\text{ind}}\big[d(X,Y)\big] := \sum_{i,j=0}^{m} d(s_i, s_j)\, p_{X;i}\, p_{Y;j}$ under the additional assumption of independence between X and Y. A measure of dependence between X and Y is then defined as

$$\kappa_d(X,Y) = \frac{E_{\text{ind}}\big[d(X,Y)\big] - E\big[d(X,Y)\big]}{E_{\text{ind}}\big[d(X,Y)\big]}, \tag{18}$$

which can be understood as a type of *weighted Cohen's* κ (Cohen 1968) with the weights being implied by the distance measure d. By construction, it takes the value 0 if X and Y are independent. If using the Hamming distance (1), $d = d_{\text{H}}$, then

$$E\big[d_{\text{H}}(X,Y)\big] = \textstyle\sum_{i,j=0}^{m} (1 - \delta_{i,j}) \cdot p_{XY;ij} = 1 - \sum_{i=0}^{m} p_{XY;ii},$$

so $\kappa_{d_{\text{H}}}(X,Y)$ becomes the unweigthed Cohen's κ and thus a measure of nominal agreement. If considering $d_{\text{o},2}$ from (3) (quadratic weighting scheme), in contrast, we get

$$E\big[d_{\text{o},2}(X,Y)\big] = E\big[(I_X - I_Y)^2\big] = E[I_X^2] + E[I_Y^2] - 2\,E[I_X\, I_Y],$$

that is, the difference $E_{\text{ind}}\big[d_{\text{o},2}(X,Y)\big] - E\big[d_{\text{o},2}(X,Y)\big]$ just equals twice the covariance of the rank counts I_X, I_Y, also see Cohen (1968). Thus, $\kappa_{d_{\text{o},2}}(X,Y)$ measures for linear dependence between the rank counts. As a final example, let us look at the ordinal block distance (2) (linear weighting scheme), then

$$E\big[d_{\text{o},1}(X,Y)\big] = E\big[|I_X - I_Y|\big] = \textstyle\sum_{i=0}^{m-1} \big(f_{X;i} + f_{Y;i} - 2\, f_{XY;ii}\big), \tag{19}$$

see Supplement A.4 for the proof. So $\kappa_{d_{\text{o},1}}(X,Y)$ is a measure of quadrant dependence in the sense of Lehmann (1966). Since this measure of dependence solely relies on cumulative probabilities, it is of particular relevance for ordinal dependence analysis.

7. Distance-Based Analysis of Ordinal Time Series

From now on, we are concerned with time series data, that is, $x_1, \ldots, x_n$ are assumed to be a time series from a stationary ordinal process $(X_t)_{\mathbb{Z}}$ (excluding again the deterministic case $\text{disp}_d(X_t) = 0$). This data-generating process (DGP) is also assumed to satisfy an appropriate mixing condition such that the application of a central limit theorem (CLT) for serially dependent data is possible, see Supplement A.5 for more details. Note that we may interpret the results of Section 4 as a special case of the upcoming ones by assuming the DGP to be iid. Actually, the proofs in Supplement A.6 already followed this strategy and therefore also apply to Section 7.1 about marginal properties of $(X_t)_{\mathbb{Z}}$. In Section 7.2, we define measures of serial dependence regarding $(X_t)_{\mathbb{Z}}$ by adapting the dependence approach of Section 6. For this purpose, we make use of the lagged bivariate probabilities

$$\begin{aligned} f_{ij}^{(h)} &= P(X_t \leq s_i, X_{t-h} \leq s_j) \\ &= P(I_t \leq i, I_{t-h} \leq j) \quad \text{for } i, j = 0, \ldots, m-1 \end{aligned}$$

as well as their corresponding sample frequencies $\hat{f}_{ij}{}^{(h)}$. Furthermore, we shall assume from now on that $(X_t)_{\mathbb{Z}}$ has some kind of "short memory" in the following sense: $\sum_{h=1}^{\infty} \big|f_{ij}^{(h)} - f_i f_j\big| < \infty$ has to hold for all $i, j = 0, \ldots, m-1$.

7.1. Marginal Properties

In this section, our focus is on the marginal distribution of $(X_t)_{\mathbb{Z}}$ such that the measures of location, dispersion, and symmetry for ordinal random variables as proposed in Section 3 are relevant. Their sample counterparts are computed in the same way as in Section 4, but since $(X_t)_{\mathbb{Z}}$ is generally serially dependent, the asymptotic distributions derived in Section 4 are not valid anymore in the time series case. As an example, the asymptotic result $\sqrt{n}\,(\hat{\boldsymbol{f}} - \boldsymbol{f}) \xrightarrow{\text{d}} \text{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$ given in (16) changes to

$$\sigma_{i,j} = f_{\min\{i,j\}} - f_i f_j + \textstyle\sum_{h=1}^{n-1} (1 - \frac{h}{n}) \big(f_{ij}^{(h)} + f_{ji}^{(h)} - 2 f_i f_j\big), \tag{20}$$

see Supplement A.5. In implementations, it might be more convenient to replace the last sum by $\sum_{h=1}^{\infty} \big(f_{ij}^{(h)} + f_{ji}^{(h)} - 2 f_i f_j\big)$, which does not make a relevant difference except for small n. This practical hint also applies to the following theorem, which constitutes a generalization of Theorem 4.1 for the case of serial dependence.

Theorem 7.1.1. Let $X_1, \ldots, X_n$ be a time series from the stationary ordinal process $(X_t)_{\mathbb{Z}}$ with range $\mathcal{S} = \{s_0, \ldots, s_m\}$ and with distance measure $d_{\text{o},1}$.

(i) If $(f_0, \ldots, f_{m-1}) \neq (\frac{1}{2}, \ldots, \frac{1}{2})$ (extreme two-point distribution), then $\widehat{\text{disp}}_{d_{\text{o},1}}$ is asymptotically normally distributed with exact mean

$$(1 - \tfrac{1}{n})\, \text{disp}_{d_{\text{o},1}} - \tfrac{4}{n} \textstyle\sum_{h=1}^{n-1} (1 - \frac{h}{n}) \sum_{i=0}^{m-1} \big(f_{ii}^{(h)} - f_i^2\big)$$

and approximate variance

$$\begin{aligned} &\tfrac{4}{n} \textstyle\sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \big(f_{\min\{i,j\}} - f_i f_j\big) \\ &\quad + \tfrac{8}{n} \textstyle\sum_{h=1}^{n-1} (1 - \frac{h}{n}) \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \big(f_{ij}^{(h)} - f_i f_j\big). \end{aligned}$$

(ii) If the distribution of the X_i is not symmetric, then $\widehat{\text{asym}}_{d_{\text{o},1}}$ is asymptotically normally distributed with exact mean

$$\begin{aligned} &\text{asym}_{d_{\text{o},1}} + \frac{1}{n}\, \text{disp}_{d_{\text{o},1}} + \frac{2}{n} \sum_{j=0}^{m-1} \big(f_{\min\{m-j-1,j\}} - f_j f_{m-j-1}\big) \\ &\quad + \frac{4}{n} \sum_{h=1}^{n-1} (1 - \frac{h}{n}) \\ &\quad \times \sum_{j=0}^{m-1} \big(f_{jj}^{(h)} - f_j^2 + f_{j,m-j-1}^{(h)} - f_j f_{m-j-1}\big) \end{aligned}$$

and approximate variance

$$\frac{16}{n}\sum_{i,j=0}^{m-1}(1-f_i-f_{m-i-1})(1-f_j-f_{m-j-1})$$
$$\times\big(f_{\min\{i,j\}}-f_if_j\big)+\frac{32}{n}\sum_{h=1}^{n-1}(1-\frac{h}{n})$$
$$\times\sum_{i,j=0}^{m-1}(1-f_i-f_{m-i-1})$$
$$\times(1-f_j-f_{m-j-1})\big(f_{ij}^{(h)}-f_if_j\big).$$

(iii) $\widehat{\text{skew}}_{d_{o,1}}$ is asymptotically normally distributed with exact mean $\text{skew}_{d_{o,1}}$ and variance

$$\frac{4}{n}\sum_{i,j=0}^{m-1}\big(f_{\min\{i,j\}}-f_if_j\big)$$
$$+\ \frac{8}{n}\sum_{h=1}^{n-1}(1-\frac{h}{n})\sum_{i,j=0}^{m-1}\big(f_{ij}^{(h)}-f_if_j\big).$$

Parts (i)–(iii) are easily adapted to the respective normalized versions $\widehat{\text{IOV}}=\frac{2}{m}\,\widehat{\text{disp}}_{d_{o,1}}$, $\frac{1}{m}\,\text{asym}_{d_{o,1}}$, and $\frac{1}{m}\,\text{skew}_{d_{o,1}}$.

The proof of Theorem 7.1.1 is provided by Supplement A.6. Note that Theorem 7.1.1 reduces to Theorem 4.1 in the case of pairwise independence, that is, if $f_{ij}{}^{(h)}=f_if_j$ for $h\geq 1$. In the case of serial dependence, in turn, we have to compute the values of the $f_{ij}{}^{(h)}$ to be able to evaluate the asymptotics of Theorem 7.1.1. For example, if $(X_t)_{\mathbb{Z}}$ is a (finite) Markov chain with transition matrix $\mathbf{P}=(p_{i|j})$ and stationary marginal distribution $\boldsymbol{p}$, then the h-step-ahead transition probabilities $p_{i|j}{}^{(h)}=P(X_t=s_i\mid X_{t-h}=s_j)$ are the entries of $\mathbf{P}^h$. So $f_{ij}{}^{(h)}$ is computed by accumulating the joint bivariate probabilities $p_{ij}{}^{(h)}=P(X_t=s_i,X_{t-h}=s_j)=p_{i|j}{}^{(h)}\,p_j$.

Remark 7.1.2. In analogy to the iid-case, see Remark 4.2, it is useful to also have an asymptotic distribution for the boundary cases excluded in parts (i) and (ii) of Theorem 7.1.1. Assertion (ii) of Remark 4.2 holds in exactly the same way also in the general stationary case, but with $\boldsymbol{\Sigma}$ now being given by (20). Assertion (i) has to be modified as follows: $n\,(\widehat{\text{IOV}}-1)$ asymptotically follows $-4\,\sigma_{0,0}\,Z^2$ with $\sigma_{0,0}=\frac{1}{4}+\frac{2}{n}\sum_{h=1}^{n-1}(1-\frac{h}{n})\,\big(f_{00}{}^{(h)}-\frac{1}{4}\big)$ according to (20).

The finite-sample performance of the asymptotics provided by Theorem 7.1.1 is investigated in Supplement C.2.

7.2. Serial Dependence

Like before, let $(X_t)_{\mathbb{Z}}$ be a stationary process with the marginal distribution being the same as that of X, where X satisfies $\text{disp}_d(X)>0$. Picking up Equation (18), we define

$$\kappa_d(h)\ =\ \frac{\text{disp}_d(X)-E\big[d(X_t,X_{t-h})\big]}{\text{disp}_d(X)}\qquad\text{for } h\geq 1 \tag{21}$$

as a measure of serial dependence at lag h. We can interpret $\text{disp}_d(X)$ from (5) as the mean of $d(X_t,X_{t-h})$ under the null hypothesis of serial independence between X_t and X_{t-h}, that is, $\kappa_d(h)$ measures the relative deviation of the diversity coefficients for dependent and independent random variables.

If $(X_t)_{\mathbb{Z}}$ is a stationary categorical process with range $\mathcal{S}$ and with $d=d_{\text{H}}$ being the Hamming distance (1), then $E\big[d_{\text{H}}(X_t,X_{t-h})\big]=1-\sum_{i=0}^m p_{ii}^{(h)}$, see Section 6, and thus

$$\kappa_{d_{\text{H}}}(h)\ =\ \frac{\sum_{i=0}^m p_{ii}^{(h)}-s_2(\boldsymbol{p})}{1-s_2(\boldsymbol{p})}. \tag{22}$$

This is the serial Cohen's $\kappa(h)$ as proposed by Weiß and Göb (2008), having the range $\big[\frac{-s_2(\boldsymbol{p})}{1-s_2(\boldsymbol{p})};1\big]$. It defines (signed) serial dependence in terms of the (dis)agreement of the lagged random variables.

If the range $\mathcal{S}$ is ordinal, and using the ordinal block distance (2) within Equation (21) (thus a linear weighting scheme), then $E\big[d_{o,1}(X_t,X_{t-h})\big]=E\big[|I_t-I_{t-h}|\big]$. Using Equations (7) and (19), we obtain

$$\kappa_{d_{o,1}}(h)\ =\ \frac{\text{disp}_{d_{o,1}}-E\big[|I_t-I_{t-h}|\big]}{\text{disp}_{d_{o,1}}}\ =\ \frac{\sum_{i=0}^{m-1}\big(f_{ii}^{(h)}-f_i^2\big)}{\sum_{i=0}^{m-1}f_i(1-f_i)}, \tag{23}$$

where the latter equality is proven in Supplement A.7. So $\kappa_{d_{o,1}}(h)$, having the range $\big[-\sum_{i=0}^{m-1}f_i^2/\sum_{i=0}^{m-1}f_i(1-f_i);1\big]$, analyzes agreement in terms of the joint bivariate CDF of X_t and X_{t-h} (quadrant dependence). The strongest degree of negative dependence is obtained if all $f_{ii}{}^{(h)}=0$, that is, the event $X_{t-h}\leq s_i$ is necessarily followed by $X_t>s_i$. Maximal positive dependence, in turn, happens if $X_{t-h}\leq s_i$ is necessarily followed by $X_t\leq s_i$.

If $(X_t)_{\mathbb{Z}}$ would be a real-valued process with variance $\sigma^2<\infty$ and autocorrelation function (ACF) $\rho(h)$, and if d would be chosen as the squared Euclidean distance, then

$$E\big[(X_t-X_{t-h})^2\big]\ =\ E\big[X_t^2+X_{t-h}^2-2X_tX_{t-h}\big]\ =\ 2\,\big(1-\rho(h)\big)\,\sigma^2.$$

So the distance-based $\kappa_d(h)$ according to (21) also contains the ordinary ACF as a special case. The ordinal counterpart, obtained by using $d_{o,2}$ from (3) (quadratic weighting scheme), is given by

$$\kappa_{d_{o,2}}(h)\ =\ \rho_I(h), \tag{24}$$

that is, the serial dependence of $(X_t)_{\mathbb{Z}}$ at lag h is equal to the lag-h autocorrelation of the corresponding rank count process $(I_t)_{\mathbb{Z}}$, also see Section 6. So $\kappa_{d_{o,2}}(h)$ measures for linear dependence between the lagged rank counts.

Given the time series data $x_1,\ldots,x_n$, a sample version $\hat{\kappa}_d(h)$ of the general distance-based $\kappa_d(h)$ from (21) is obtained by using the sample version $\widehat{\text{disp}}_d$ of disp_d from (5) on the one hand, and $\frac{1}{n-h}\sum_{t=h+1}^n d(x_t,x_{t-h})$ for $E\big[d(X_t,X_{t-h})\big]$ on the other hand. Simplified sample versions exist for the special cases of Equations (22)–(24). For $\kappa_{d_{\text{H}}}(h)$ from (22), one replaces the (bivariate) probabilities by their corresponding sample frequencies; see Weiß (2011, 2013) for the stochastic properties of $\hat{\kappa}_{d_{\text{H}}}(h)$. Analogously, it recommends itself to use the sample cumulative frequencies $\hat{f}_i,\hat{f}_{ii}{}^{(h)}$ for defining a sample version of $\kappa_{d_{o,1}}(h)$ in (23). Finally, it is reasonable to use the sample version $\hat{\rho}_I(h)$ of $\rho_I(h)$ as the sample version of $\kappa_{d_{o,2}}(h)$ in (24).

If being interested in testing for significant dependence, it suffices to consider the distribution of these sample versions under the null of serial independence. This is exemplified by the following theorem, which considers the sample measure $\hat{\kappa}_{d_{o,1}}(h)$ if being computed from iid data, in which case $\kappa_{d_{o,1}}(h) = 0$ holds for all $h \geq 1$.

Theorem 7.2.1. Let $X_1, \ldots, X_n$ be a time series from the iid ordinal process $(X_t)_{\mathbb{Z}}$ with range $\mathcal{S} = \{s_0, \ldots, s_m\}$ and distance measure $d_{o,1}$. Let $h \geq 1$.

Then an asymptotic approximation for the distribution of $\hat{\kappa}_{d_{o,1}}(h)$ is the normal distribution with mean $-1/n$ and variance

$$\frac{1}{n}\,\frac{4}{\mathrm{disp}^2_{d_{o,1}}}\,\sum_{k,l=0}^{m-1}\big(f_{\min\{k,l\}} - f_k f_l\big)^2.$$

The proof of Theorem 7.2.1 is presented in Supplement A.7. Note that the bias correction $-1/n$ also applies to $\hat{\kappa}_{d_{\mathrm{H}}}(h)$ in the iid-case, see Weiß (2011), and it also holds for the sample ACF $\hat{\rho}_I(h)$ according to Kendall (1954).

Some simulation results concerning $\hat{\kappa}_{d_{o,1}}(1)$ are presented in Supplement C.3.

8. Application: Credit Ratings

Trading Economics (TE) is an online platform that offers a wide variety of economic data for 196 countries, including current and historical credit ratings, see *https://tradingeconomics.com/country-list/rating*. In what follows, we concentrate on the 28 countries of the European Union (EU) and consider their credit ratings according to Standard & Poor (S&P). The possible outcomes range from "D" (worst) to "AAA" (best), where ratings from "CCC" to "AA" are commonly refined by adding a plus or minus sign. So altogether, the refined range consists of the $m + 1 = 23$ states $s_0, \ldots, s_{22}$ given by "D," "SD," "R," "CC," "CCC−," "CCC," "CCC+," "B−," "B," "B+," "BB−," "BB," "BB+," "BBB−," "BBB," "BBB+," "A−," "A," "A+," "AA−," "AA," "AA+," "AAA." Motivated by the TE credit rating, we treat the refined states $s_0, \ldots, s_{22}$ as equidistant, so the distance measure $d_{o,1}$ is used for corresponding analyses.

From the historical information provided by Trading Economics, monthly time series of S&P credit ratings per country were generated, where the rating in month t was defined to be the latest rating in that month. The available periods of ratings differ between countries; to have a unique time range for each of the EU countries, we restrict to the $n = 216$ months from January 2000 to December 2017.

The 28 time series show quite different shapes, ranging from constant time series to such having abrupt up and down movements (financial crisis). Figure 1 shows an example series with moderate fluctuations, the S&P credit ratings of Latvia. Its normalized dispersion takes the low value 0.155, and it has asymmetry 0.240 and skewness −0.318. In Figure 1, we also see a plot of the serial dependence measure $\hat{\kappa}_{d_{o,1}}(k)$ against time lag k; obviously, we are concerned with a strong and slowly decaying positive dependence. This is reasonable in view of the time series plot with its long runs and small jumps of values. Also for the remaining (nonconstant) time series, we have such a strong positive dependence. Note that the plotted critical values (5 % level; dashed lines) are computed according to the normal approximation of Theorem 7.2.1 by plugging-in the sample CDF.

Figure 2 plots marginal properties of the 28 EU countries' S&P credit rating time series against each other. For all countries, the sample median (i.e., $\hat{x}_{\mathrm{loc},d_{o,1}}$ according to (4) in Section 3.1) is at least $s_{12} = $ BB+, and the normalized dispersion measure $\frac{2}{m}\,\widehat{\mathrm{disp}}_{d_{o,1}}$ takes values between 0 and 0.5. Note that zero dispersion is observed for Germany and Luxembourg since these two time series are constantly $s_{22} = $ AAA. A very low dispersion (say, <0.1) is also observed for some countries with median lower than s_{22}, always caused by nearly constant time series.

The plot of $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$ against the median in Figure 2 shows that low asymmetry generally goes along with a central median, whereas a large median goes along with large asymmetry. It should be noted, however, that there is still a lot of variation around this general pattern, so $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$ provides additional information compared to the median. In the plot of $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$ against $\frac{2}{m}\,\widehat{\mathrm{disp}}_{d_{o,1}}$, all points are plotted below the anti-diagonal (with a lot of fluctuation), which is reasonable because of the inequality $\frac{1}{m}\,\mathrm{asym}_{d_{o,1}} \leq 1 - \mathrm{IOV}$, see Section 3.3. In the plot of $\frac{1}{m}\,\widehat{\mathrm{skew}}_{d_{o,1}}$ against $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$, in turn, the points follow closely a linear relation, where skewness is always negative (most of the probability mass accumulates in the upper half of the range). Also note that we have some kind of clustering in all $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}}$-plots because of a gap between 0.65 and 0.85: the countries with $\frac{1}{m}\,\widehat{\mathrm{asym}}_{d_{o,1}} > 0.85$ never had a ranking worse than AA.

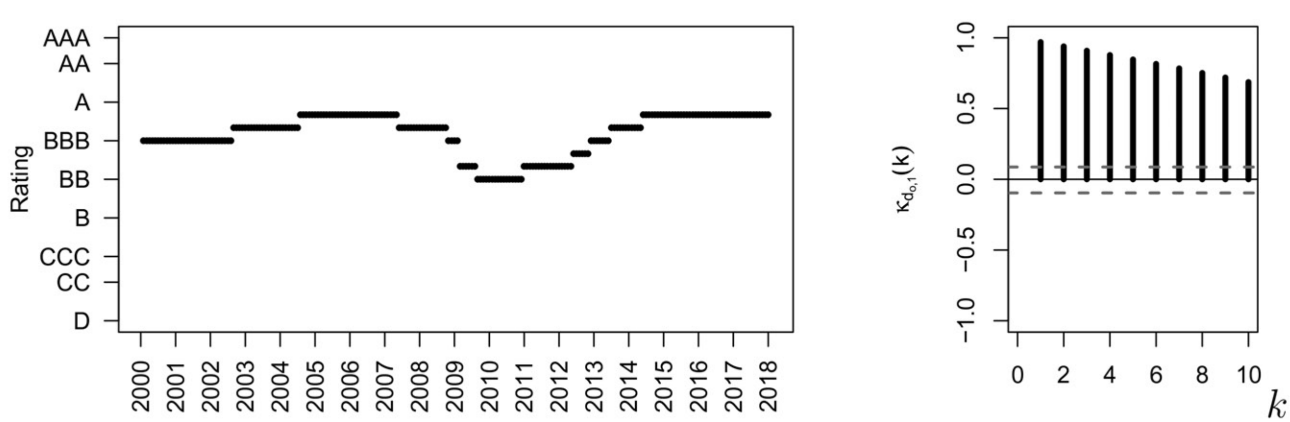

Figure 1. Monthly S&P credit rating of Latvia: time series plot (left), and plot of $\hat{\kappa}_{d_{o,1}}(k)$ against time lag k (right).

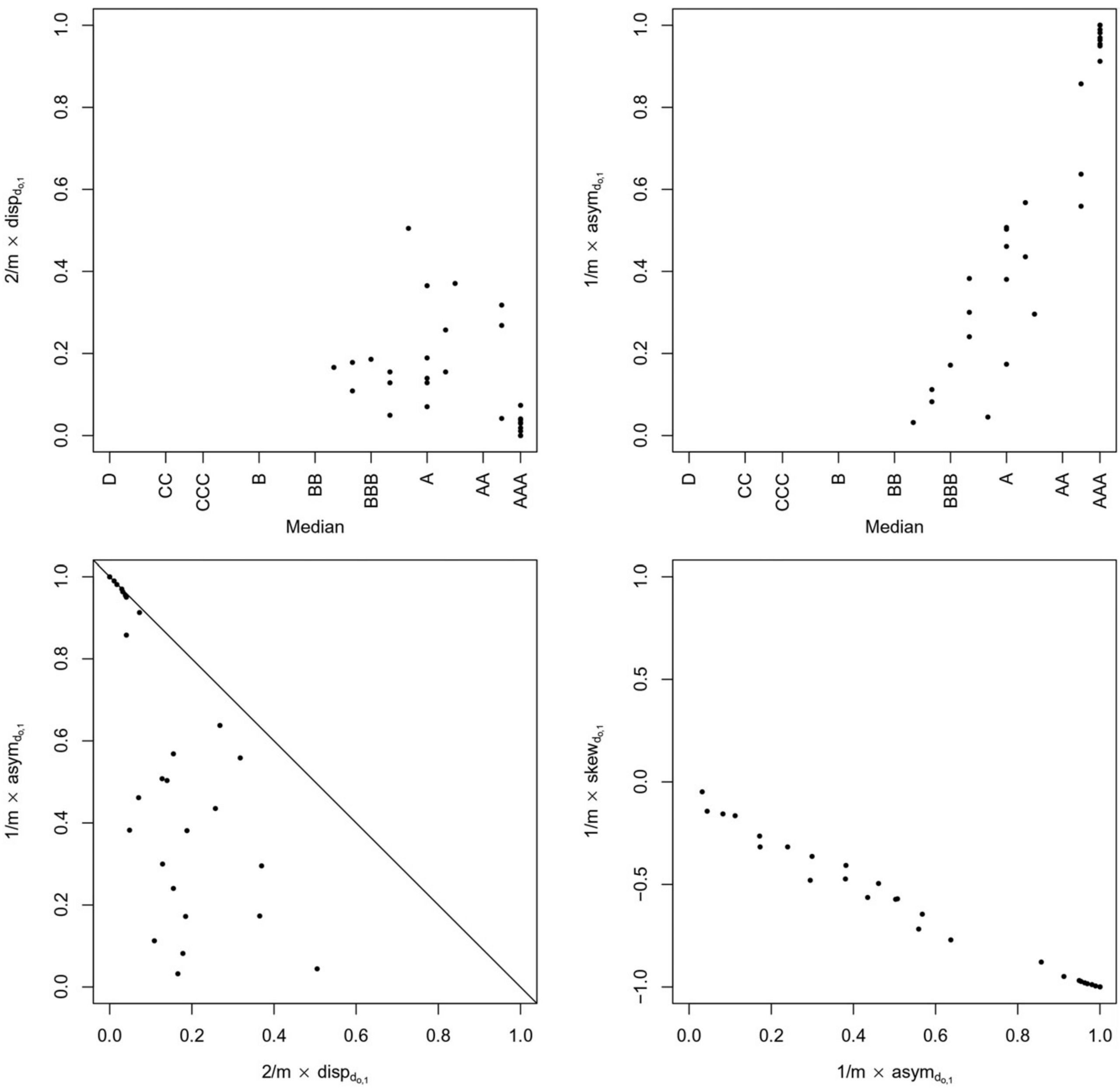

Figure 2. Marginal properties of S&P credit rating time series of 28 EU countries.

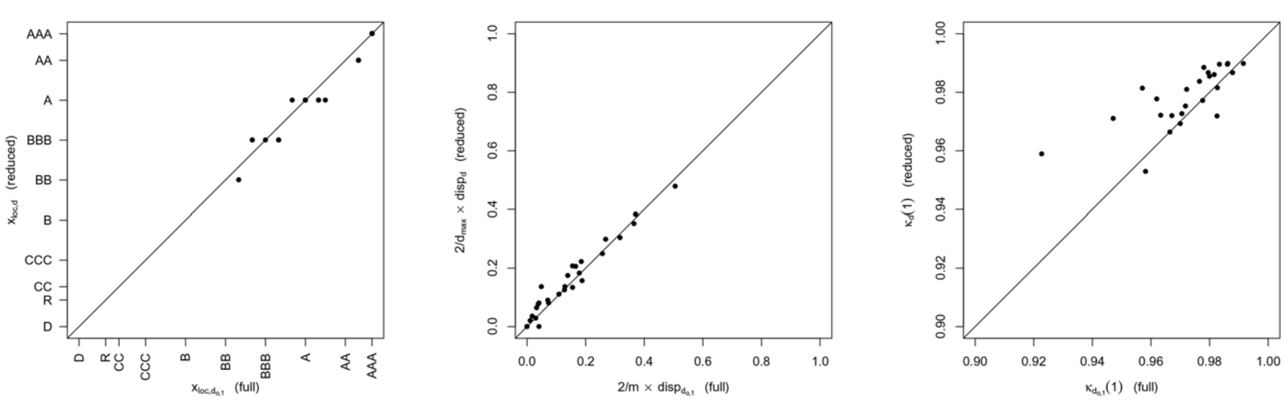

Figure 3. Comparison of properties of S&P credit rating time series of 28 EU countries if using either full range or reduced range.

Finally, let us return to the definition of the ordinal range. Up to now, we considered the refined range consisting of 23 states, which we treated as equidistant. It would also have been reasonable to consider the reduced range with the $\tilde{m}+1=10$ states $\tilde{s}_0,\ldots,\tilde{s}_9$ given by "D," "R," "CC," "CCC," "B," "BB," "BBB," "A," "AA," "AAA." But then, to make results comparable and again in view of the TE rating, it would not be appropriate anymore to treat these states as equidistant. Therefore, the following (additive) distance matrix is used in case of the reduced range:

$$\mathbf{D} = \big(d(\tilde{s}_i,\tilde{s}_j)\big)_{i,j=0,\ldots,9} = \begin{pmatrix} 0 & 2 & 3 & 5 & 8 & 11 & 14 & 17 & 20 & 22 \\ 2 & 0 & 1 & 3 & 6 & 9 & 12 & 15 & 18 & 20 \\ 3 & 1 & 0 & 2 & 5 & 8 & 11 & 14 & 17 & 19 \\ 5 & 3 & 2 & 0 & 3 & 6 & 9 & 12 & 15 & 17 \\ 8 & 6 & 5 & 3 & 0 & 3 & 6 & 9 & 12 & 14 \\ 11 & 9 & 8 & 6 & 3 & 0 & 3 & 6 & 9 & 11 \\ 14 & 12 & 11 & 9 & 6 & 3 & 0 & 3 & 6 & 8 \\ 17 & 15 & 14 & 12 & 9 & 6 & 3 & 0 & 3 & 5 \\ 20 & 18 & 17 & 15 & 12 & 9 & 6 & 3 & 0 & 2 \\ 22 & 20 & 19 & 17 & 14 & 11 & 8 & 5 & 2 & 0 \end{pmatrix}.$$

It should be noted that this distance is not centrosymmetric (e.g., $d(\tilde{s}_0,\tilde{s}_2)=3\neq 5=d(\tilde{s}_9,\tilde{s}_7)$), that is, the symmetry approaches discussed in Section 3.3 cannot be applied here. The remaining measures of location, dispersion, and serial dependence are still available, although they now have to be computed with respect to d (see Sections 4 and 7.2): location $\hat{x}_{\mathrm{loc},d} = \arg\min_x \frac{1}{n}\sum_{t=1}^{n} d(x_t,x)$ instead of the sample median $\hat{x}_{\mathrm{loc},d_{\mathrm{o},1}}$, and similarly dispersion $\widehat{\mathrm{disp}}_d = \hat{\boldsymbol{p}}^\top\mathbf{D}\hat{\boldsymbol{p}}$ and serial dependence $\hat{\kappa}_d(h) = \big(\widehat{\mathrm{disp}}_d - \frac{1}{n-h}\sum_{t=h+1}^{n} d(x_t,x_{t-h})\big)\big/\widehat{\mathrm{disp}}_d$. Note that $\widehat{\mathrm{disp}}_d$ has to be normalized with the factor $2/d(\tilde{s}_0,\tilde{s}_9)=1/11$ because of Theorem A.2.2 in the supplement.

Now the crucial question is: how do location, dispersion, and serial dependence change if using the reduced range instead of the full one? This is analyzed in Figure 3. It can be seen that location and dispersion do not change a lot, so these characteristics are robust against this change in range. The serial dependence values in the right graph of Figure 3, in contrast, are stronger affected, with mainly larger values in the reduced case. This is reasonable, because the reduction of states causes less variation, that is, runs tend to become longer thus leading to increased positive dependence. As an example, the Slovenia time series switched several times between A− and AA in the full-range case, but now only twice between A and AA. Finally, note that because of the noncentrosymmetric distance, the dispersion of the original time series and reflected copies thereof might differ. Considering, again the Latvia time series, for example, then the original time series has the normalized sample dispersion 0.207, which changes after reflection (replacing $\tilde{s}_i$ by $\tilde{s}_{9-i}$) to 0.165.

9. Conclusions

We discussed measures of location, dispersion, symmetry, and (serial) dependence based on expected distances for the case of ordinal (time series) data. These measures are well interpretable and can deal with many types of distance function defined on the given range. Also sample counterparts to these measures as well as stochastic properties thereof have been analyzed, thus allowing for statistical inference regarding ordinal data. Two real applications demonstrated the usefulness of these distance-based measures.

For future research, certainly further analyses regarding ordinal time series should be considered, for example, concerning the asymptotics of alternative ordinal variation measures like those proposed in Kvålseth (2011a). In particular, it should be emphasized that the distance-based approach can also be used for random variables and processes having other types of range, for example, compositional data. It thus allows to define novel analytic tools for such data types, and it offers the opportunity to treat different data types with a unique approach and a unique interpretation.

Supplementary Materials

Supplement A: Proofs of the results in the article "Distance-Based Analysis of Ordinal Data and Ordinal Time Series".
Supplement B: Numerical illustrations regarding Section 3.
Supplement C: Results from a simulation study regarding the sample measures discussed in Sections 4 and 7.
Supplement D: Overview of some distance-based ordinal measures.

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